

Technical Document #339: Computer Vision - Comprehensive Overview

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Semantic segmentation classifies each pixel in an image into a category. It is used in autonomous driving and medical image analysis.

Video understanding analyzes temporal dynamics in video sequences. 3D CNNs and transformer-based models capture spatiotemporal features.

Image super-resolution reconstructs high-resolution images from low-resolution inputs. GAN-based methods produce photorealistic upscaled images.

Optical character recognition (OCR) converts images of text into machine-readable text. It is used in document digitization and automation.

Image generation creates new images from random noise or text descriptions. Diffusion models like Stable Diffusion produce photorealistic images.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Image classification assigns a label to an image from a fixed set of categories.
- Optical character recognition (OCR) converts images of text into machine-readable text.
- Image generation creates new images from random noise or text descriptions.